# Hesamoddin Haddadadel

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## Research Summary

I build language-driven robotic systems that run entirely on local hardware. At GIST, I designed a privacy-preserving, fully local LLM assistant for rehabilitation robotics (ICROS 2026), and my M.Sc. thesis extends this toward vision–language–action (VLA) models that produce both verbal and physical outputs on edge-deployed robots. I bring a control-engineering background, a first-author IEEE publication on learning-based prosthesis control, and industry experience deploying and operating LLMs in production.

## Education

**Master of Science: Mechanical and Robotics Engineering (candidate)** 09/2025 – Present

**Gwangju Institute of Science and Technology (GIST)**, Gwangju, South Korea
HUR Group (BioMechatronics Lab); Advisor: Prof. Pilwon Hur
M.Sc. Thesis (in progress): **Vision–language–action models with joint verbal and action outputs for edge-deployed robotic systems**

**Bachelor of Science: Mechanical Engineering** 2018 – 2023

**University of Tehran**, Tehran, Iran Subdiscipline: Control Engineering
B.Sc. Thesis (92.5/100): **Knee & femur joint angle estimation based on mechanical & electromyographic measurements for active control of transfemoral prostheses**

## Research Interests

Vision–language–action (VLA) models, physical AI, multimodal learning, robot learning and reinforcement learning, edge deployment of foundation models, human–robot interaction, learning-based control

## Publications

**H. Haddad** and P. Hur, "A Privacy-Preserving Local RAG System for Rehabilitation Knowledge Assistance," *41st ICROS Annual Conference*, Daegu, pp. 199–200, July 2026.

**H. Haddadadel**, A. Yousefi-Koma, and M. Asgharzadeh, "Knee & femur joint angle estimation based on mechanical & electromyographic measurements for active control of transfemoral prostheses," *2023 11th RSI International Conference on Robotics and Mechatronics (ICRoM)*, IEEE, Dec. 2023. [DOI]

## Research Experience

**HUR Group (BioMechatronics Lab), GIST** 09/2025 – Present

Advisor: Prof. Pilwon Hur
**Project:** Language interfaces for upper-limb rehabilitation robots

- Designed and built a fully local retrieval-augmented generation (RAG) assistant for rehabilitation therapists and patients, satisfying patient-privacy and closed hospital-network constraints (ICROS 2026).
- Built the retrieval stack end to end: a versioned ingestion pipeline over clinical reference material, hybrid dense and full-text retrieval, and evaluation of open-weight generators for edge deployment.
- Integrating the assistant with an audio interface on the rehabilitation robot; own the MLOps and service integration across the system.
- Set up and administer the lab's self-hosted infrastructure for ∼30 users: Mattermost, Forgejo, Wiki.js and a custom lab application running in Docker on a Synology NAS, with RAID storage and security hardening that passed the university's network-security review.

**Center of Advanced Systems and Technologies (CAST), University of Tehran** 05/2023 – 09/2023

Supervisor: Prof. Aghil Yousefi-Koma
**Project:** Joint angle estimation for active transfemoral prostheses

- Developed a lightweight multilayer neural network estimating knee and femur joint angles from mechanical and electromyographic signals, as the reference input for active prosthesis control; led to a first-author IEEE publication (ICRoM 2023).
- Handled sensor logging, data analysis, and feature extraction from mechanical and EMG measurements.

**University of Tehran in collaboration with Taleghani Hospital** 07/2023 – 11/2023

Supervisor: Dr. Mohammad Khoshnevisan

**Project:** Clinical grading of fatty liver disease

- Built an ensemble of deep neural networks, gradient-boosted decision trees, and statistical models to grade fatty liver disease from blood-test, demographic, and anthropometric data.

**University of Tehran, Department of Mechanical Engineering** 2022

Supervisor: Dr. Mohammad Khoshnevisan

Designed neural network regressors for the highly non-linear mapping from alloy composition to mechanical properties.

## Industry Experience

**Software Developer, RavanErtebat Co.** 02/2024 – 05/2025

- Deployed and maintained transformer-based and large language models in production back-end services.
- Handled Kubernetes administration and infrastructure design as part of DevOps responsibilities.

**Machine Vision Intern, Irankhodro Powertrain Co.** Summer 2022

- Designed and deployed models for live object detection within a team developing autonomous vehicle features.

## Teaching Experience

**Teaching Assistant – Artificial Intelligence, University of Tehran** 09/2022 – 07/2023

Lecturer: Dr. Mohammad Khoshnevisan. English-language course, 50+ students. Taught linear algebra, neural networks, CNNs, reinforcement learning, unsupervised learning, and decision trees.

**Teaching Assistant – Dynamics, University of Tehran** 06/2022 – 09/2022

Lecturer: Dr. Farzad Ayatollahzadeh Shirazi

**English Teacher, Parseek English Institute** 2022

200+ hours teaching English to children and teenagers.

## Skills

**Machine Learning:** PyTorch, TensorFlow, transformers and LLMs, retrieval-augmented generation, local/edge LLM deployment (Ollama)
**MLOps & Infrastructure:** Kubernetes, FastAPI, Qdrant, MongoDB, multi-service integration
**Programming:** Python, C++, Go, C#, MATLAB, LaTeX

## Honors

**National University Entrance Exam (Konkour)** 2018

Ranked 315$^{th}$ among 144,000 participants (top 0.2%).

**IELTS Academic** February 2023

Overall band score: 8.0

**Top Undergraduate Grades** 2023

Statistics and Probability (95/100), Artificial Intelligence (93.75/100), Calculus 2 (91.25/100), Engineering Mathematics (90/100)

## References

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